

KINETICS & REACTOR DESIGN

CHE F311 Problems and Solutions I-Semester 2026-27

Every numerical question in the uploaded decks, with its solution.

Every question below is quoted from one of the uploaded files, and every solution uses the method taught in the deck that set it. The questions are grouped by source deck (Parts A–I). Where a slide prints its own answer the solution reproduces it; where a slide leaves something undefined, the Notes box says what was assumed. All numbers are re-computed by `krd-answers-check.py` kept beside this file.

Question inventory

ID	Source	Topic	Solved at
Q1	Ch1, slide 36	Rocket engine: rates of reaction from mole balance	p. ??
Q2	Ch1, slide 54	$A \rightarrow B$: CSTR, PFR and batch volumes/times, three rate laws	p. ??
Q3	Ch2, slides 19–24	Levenspiel sizing: CSTR and PFR at 80%	p. ??
Q4	Ch2, slides 31–36	Two CSTRs in series (40%, 80%)	p. ??
Q5	Ch2, slides 37–38	Two PFRs in series	p. ??
Q6	Ch2, slides 41–42	Mixed reactor, unknown stoichiometry	p. ??
Q7	Ch2, slides 45–46	PFRs in parallel branches: feed split	p. ??
Q8	Ch2, slide 59	P2-7 adiabatic $A + B \rightarrow C$ data, (a)–(e)	p. ??
Q9	Ch3, slides 5–7	Relative rates: $2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$	p. ??
Q10	Ch3, slides 6–7	Relative rates: $2A + 3B \rightarrow 5C$	p. ??
Q11	Ch3, slides 42–44	Saponification stoichiometric table	p. ??
Q12	Ch3, slides 48–51	Flow-system stoichiometric table, $A + 2B \rightarrow C + D$	p. ??
Q13	Ch3 s.61 (+Ch5 s.77)	Dimerisation $2A \rightarrow R$ in MFR: rate equation	p. ??
Q14	Ch3, slide 63	Phosphine cracking: PFR size	p. ??
Q15	Ch4, slides 17–20	Ethylene-oxide batch data: k and batch time	p. ??
Q16	Ch4, slides 29–38	Ethylene glycol: CSTR volume, parallel and series 800-gal reactors	p. ??
Q17	Ch4, slides 44–45	Ethane cracking: PFR volume and number of tubes	p. ??
Q18	Ch4, slides 55–58	Gas $A + B \rightarrow 2C$ in PBR with pressure drop	p. ??
Q19	Ch4, slides 76–77	PBR + CSTR in series: order, exit pressure, particle size	p. ??
Q20	Ch4, slides 78–79	Series reactions $A \rightarrow B \rightarrow C$ in a batch reactor	p. ??
Q21	Ch4, slides 90–98	PFR with pressure drop: pipe length for 10%/20%	p. ??
Q22	Ch5, slides 12–17	Trityl + methanol: order and rate constant	p. ??
Q23	Ch5, slide 75	Dimethyl ether bomb: order and k	p. ??
Q24	Ch5, slide 72	First-order reversible $A \rightleftharpoons R$	p. ??
Q25	Ch5, slide 73	Half-order batch; $2A \rightarrow B$ volume change	p. ??
Q26	Ch5, slide 74	$A \rightarrow 3R$ in MFR: rate equation	p. ??
Q27	Ch5, slide 76	$A \rightarrow 2.5R$ batch, partial-pressure data	p. ??
Q28	Ch5, slides 78–79	Reaction bomb $2A \rightarrow B$ at 100 C	p. ??
Q29	Ch5, slide 80	Gambling tables (first-order analogue)	p. ??
Q30	Ch5, slide 81	$-r_A = kC_A^2$: $6 \times$ MFR and same-size PFR	p. ??
Q31	Ch5, slide 82	$A + 3B \rightarrow 6R$: X_A, X_B, C_B	p. ??
Q32	Ch5, slide 83	Gaseous feed with T, π change: $A + B \rightarrow 5R$	p. ??
Q33	Ch5, slide 84	Sucrose–sucrase Michaelis–Menten fit	p. ??
Q34	Ch5, slide 61	Reversible PFR: equilibrium and actual conversion	p. ??
Q35	Ch5, slide 62	$A \rightarrow R + 2S$ from total-pressure data	p. ??
Q36	Ch6, slide 32	Product distribution, 90% conversion, three contacting patterns	p. ??
Q37	Ch6, slide 33	Two MFRs in series, $r_R = k_1 C_A^2$, $r_S = k_2 C_A$	p. ??
Q38	Ch7, slides 14–17	n-Butane isomerisation: adiabatic PFR volume	p. ??
Q39	Ch7, slides 48–51	Adiabatic equilibrium + two interstage coolers	p. ??
Q40	Ch7, slides 57–59	Propylene-oxide CSTR operated adiabatically	p. ??
Q41	Ch7, slides 60–74	The same CSTR with a cooling coil	p. ??
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Q43	Catalysis, slide 48	Pellet concentration profile; diameter for $\eta = 0.8$	p. ??
Q44	Catalysis, slide 57	Internal diffusion: rates at 100/150 C, apparent E	p. ??
Q45	Non-ideal, s.17–19	Tracer pulse: $E(t)$ and fraction between 3 and 6 min	p. ??
Q46	Non-ideal, s.52–59	Dispersion model: conversion four ways	p. ??

The decks are photographs of slides; parts of some questions sit inside the pictures and are quoted as printed. Decks that repeat a question (the dimerisation of Q13 reappears in Ch5) are solved once and cross-referenced.

Part A – Mole balances and rates (Ch1-krd (1).pdf)

QQ1 – Rocket engine: rates of reaction. Ch1, slide 36

Source: Ch1-krd (1).pdf, slide 36 (the “Example” slide).

A rocket engine burns a stoichiometric mixture of fuel (liquid hydrogen) in oxidant (liquid oxygen). The combustion chamber is cylindrical, 75 cm long and 60 cm in diameter, and the combustion process produces 108 kg/s of exhaust gases. If combustion is complete, find the rate of reaction of hydrogen and of oxygen.

What “rate of reaction” means here. The rate is defined per unit extensive quantity of the system (Ch1 slides 30–34); for a homogeneous gas-phase combustion chamber the natural basis is the reactor *volume*, $-r_i = (\text{moles of } i \text{ reacting per unit time})/V$, with units $\text{mol}/(\text{m}^3 \cdot \text{s})$. So the whole problem is: find the molar rates at which H_2 and O_2 are consumed, and divide by the chamber volume.

Step 1 — the chamber volume. A cylinder of length $L = 0.75 \text{ m}$ and diameter $D = 0.60 \text{ m}$:

$$V = \frac{\pi}{4} D^2 L = \frac{\pi}{4} (0.60)^2 (0.75) = 0.2121 \text{ m}^3.$$

Step 2 — turn the exhaust mass flow into a molar flow. Complete combustion means every gram of exhaust leaves as water, $\text{H}_2 + \frac{1}{2} \text{O}_2 \rightarrow \text{H}_2\text{O}$, of molar mass 18 kg/kmol :

$$F_{\text{H}_2\text{O}} = \frac{108 \text{ kg/s}}{18 \text{ kg/kmol}} = 6.0 \text{ kmol/s}.$$

Step 3 — read the reactant rates off the stoichiometry. One mole of H_2 and half a mole of O_2 disappear per mole of water formed, so

$$F_{\text{H}_2, \text{reacted}} = 6.0 \text{ kmol/s}, \quad F_{\text{O}_2, \text{reacted}} = \frac{1}{2} (6.0) = 3.0 \text{ kmol/s}.$$

Step 4 — divide by the volume. At steady state with complete, uniform combustion the mole balance $\text{In} - \text{Out} + \text{Generation} = \text{Accumulation} (= 0)$ gives the consumption rate per unit volume directly:

$$-r_{\text{H}_2} = \frac{6.0}{0.2121} = 28.3 \text{ kmol m}^{-3} \text{ s}^{-1}, \quad -r_{\text{O}_2} = \frac{3.0}{0.2121} = 14.1 \text{ kmol m}^{-3} \text{ s}^{-1}.$$

Sanity checks for the exam: the two rates are in the same 2:1 ratio as the stoichiometric coefficients (this is the relative-rate rule of Part C), and the magnitudes sit in the “rocket engines” band of the rate spectrum printed on slide 35 — combustion is among the fastest reactions there are.

$$-r_{\text{H}_2} \approx 28.3 \text{ kmol}/(\text{m}^3 \cdot \text{s}) \approx 2.8 \times 10^4 \text{ mol}/(\text{m}^3 \cdot \text{s}); \quad -r_{\text{O}_2} \approx 14.1 \text{ kmol}/(\text{m}^3 \cdot \text{s}) \approx 1.4 \times 10^4 \text{ mol}/(\text{m}^3 \cdot \text{s}).$$

QQ2 – CSTR, PFR and batch for three rate laws. Ch1, slide 54

Source: Ch1-krd (1).pdf, slide 54 (the “Problem” slide).

The reaction $A \rightarrow B$ is to be carried out isothermally in a continuous-flow reactor. Calculate both the CSTR and the PFR volumes necessary to consume 99% of A ($C_A = 0.01 C_{A0}$) when the entering molar flow rate is $F_{A0} = 5 \text{ mol/h}$, assuming the rate laws (a) $-r_A = k$ with $k = 0.05 \text{ mol}/(\text{h} \cdot \text{dm}^3)$, (b) $-r_A = k C_A$ with $k = 0.0001 \text{ s}^{-1}$, (c) $-r_A = k C_A^2$ with $k = 3 \text{ dm}^3/(\text{mol} \cdot \text{h})$. The entering volumetric flow rate is $v_0 = 10 \text{ dm}^3/\text{h}$. (d) Repeat each case for a batch reactor reaching 99%.

Set-up used for every part. $X = 0.99$ and $C_{A0} = F_{A0}/v_0 = 5/10 = 0.5 \text{ mol/dm}^3$. The three design equations (slides 42–53):

$$\text{CSTR: } V = \frac{F_{A0} X}{(-r_A)_{\text{exit}}}, \quad \text{PFR: } V = F_{A0} \int_0^X \frac{dX}{-r_A}, \quad \text{batch: } t = N_{A0} \int_0^X \frac{dX}{-r_A V} = C_{A0} \int_0^X \frac{dX}{-r_A} \quad (V \text{ const}).$$

For a flow reactor with $-r_A = kC_A^n$ and $C_A = C_{A0}(1 - X)$, the integrals below are the standard ones; write them out in the exam rather than quoting results.

(a) Zero order, $-r_A = k = 0.05$. The rate is constant, so the CSTR exit rate equals the rate everywhere in the PFR and the two reactors coincide:

$$V_{\text{CSTR}} = \frac{F_{A0} X}{k} = \frac{5(0.99)}{0.05} = 99 \text{ dm}^3, \quad V_{\text{PFR}} = \frac{F_{A0}}{k} \int_0^{0.99} dX = 99 \text{ dm}^3.$$

$$\text{Batch: } t = \frac{C_{A0}}{k} \int_0^{0.99} dX = \frac{0.5(0.99)}{0.05} = 9.9 \text{ h}.$$

(b) First order, $-r_A = kC_A$. First fix the units: $k = 0.0001 \text{ s}^{-1} \times 3600 = 0.36 \text{ h}^{-1}$ — the single most common slip in this question. With $C_A = C_{A0}(1 - X)$:

$$\begin{aligned} V_{\text{CSTR}} &= \frac{F_{A0} X}{kC_{A0}(1 - X)} = \frac{v_0 X}{k(1 - X)} = \frac{10(0.99)}{0.36(0.01)} = 2750 \text{ dm}^3, \\ V_{\text{PFR}} &= \frac{F_{A0}}{kC_{A0}} \int_0^X \frac{dX}{1 - X} = \frac{v_0}{k} \ln \frac{1}{1 - X} = \frac{10}{0.36} (4.605) = 127.9 \text{ dm}^3, \\ t_{\text{batch}} &= \frac{1}{k} \ln \frac{1}{1 - X} = \frac{4.605}{0.36} = 12.8 \text{ h}. \end{aligned}$$

(c) Second order, $-r_A = kC_A^2 = kC_{A0}^2(1 - X)^2$.

$$\begin{aligned} V_{\text{CSTR}} &= \frac{F_{A0} X}{kC_{A0}^2(1 - X)^2} = \frac{v_0 X}{kC_{A0}(1 - X)^2} = \frac{10(0.99)}{3(0.5)(0.01)^2} = 66\,000 \text{ dm}^3, \\ V_{\text{PFR}} &= \frac{v_0}{kC_{A0}} \int_0^X \frac{dX}{(1 - X)^2} = \frac{v_0}{kC_{A0}} \cdot \frac{X}{1 - X} = \frac{10}{1.5} (99) = 660 \text{ dm}^3, \\ t_{\text{batch}} &= \frac{1}{kC_{A0}} \cdot \frac{X}{1 - X} = \frac{99}{1.5} = 66 \text{ h}. \end{aligned}$$

What the numbers teach (and what the deck's next section turns into the Levenspiel plot): as the order rises the CSTR volume explodes — $99 \rightarrow 2750 \rightarrow 66\,000 \text{ dm}^3$ — while the PFR grows mildly — $99 \rightarrow 128 \rightarrow 660 \text{ dm}^3$. The CSTR operates at the *exit* rate, i.e. at the smallest rate in the reactor, and punishing a small rate with a big volume is exactly what the design equations do. At 99% conversion a CSTR is 100× the PFR for the second-order case.

(a) 99 dm^3 (CSTR = PFR), batch 9.9 h; (b) CSTR 2750, PFR 127.9 dm^3 , batch 12.8 h; (c) CSTR 66 000, PFR 660 dm^3 , batch 66 h.

Part B – Conversion and reactor sizing (Ch2-krd (1).pdf)

QQ3 – Sizing from the Levenspiel table. Ch2, slides 19–24

Source: Ch2-krd (1).pdf, slides 19–24 (the worked “CSTR/PFR sizing” problem).

Consider the isothermal gas-phase isomerization $A \rightarrow B$ ($T = 500 \text{ K}$, $P = 830 \text{ kPa} = 8.2 \text{ atm}$, charge pure A). The measured rates are:

X	0	0.1	0.2	0.4	0.6	0.7	0.8
$-r_A \text{ (mol/m}^3\cdot\text{s)}$	0.45	0.37	0.30	0.195	0.113	0.079	0.05

Calculate the volumes required to achieve 80% conversion in a CSTR and in a PFR when species A enters at $F_{A0} = 0.4 \text{ mol/s}$.

Step 0 — build the Levenspiel column. The deck's method (slides 10–12) sizes reactors from the plot of $F_{A0}/(-r_A)$ vs X : the CSTR volume is a *rectangle* (height at exit conversion times width X); the PFR volume is the *area under the curve*. Dividing $F_{A0} = 0.4$ mol/s by each rate gives

X	0	0.1	0.2	0.4	0.6	0.7	0.8
$F_{A0}/(-r_A)$ (m ³)	0.89	1.08	1.33	2.05	3.54	5.06	8.00

Step 1 — CSTR volume (slide 20). A CSTR operates at its *exit* conditions, so evaluate the rate at $X = 0.8$: $(1/-r_A)_{0.8} = 1/0.05 = 20$ m³·s/mol, i.e. $F_{A0}/(-r_A) = 8.00$ m³, and

$$V_{\text{CSTR}} = \left(\frac{F_{A0}}{-r_A} \right)_{X=0.8} X = 0.4 \frac{\text{mol}}{\text{s}} \times 20 \frac{\text{m}^3 \cdot \text{s}}{\text{mol}} \times 0.8 = 6.4 \text{ m}^3.$$

Step 2 — PFR volume by Simpson's rule (slides 21–23). $V_{\text{PFR}} = \int_0^{0.8} \frac{F_{A0}}{-r_A} dX$ has no closed form without a rate law, so the deck applies its five-point quadrature with $h = \Delta X = (0.8 - 0)/4 = 0.2$:

$$\begin{aligned} V &= \frac{\Delta X}{3} \left[\frac{F_{A0}}{-r_A}(0) + 4 \frac{F_{A0}}{-r_A}(0.2) + 2 \frac{F_{A0}}{-r_A}(0.4) + 4 \frac{F_{A0}}{-r_A}(0.6) + \frac{F_{A0}}{-r_A}(0.8) \right] \\ &= \frac{0.2}{3} [0.89 + 4(1.33) + 2(2.05) + 4(3.54) + 8.00] = \frac{0.2}{3} (32.47) = 2.165 \text{ m}^3. \end{aligned}$$

Step 3 — read the picture (slide 24). The CSTR rectangle minus the PFR area is $6.4 - 2.165 = 4.235$ m³ of extra volume — the price of running the whole reactor at the slow exit rate.

Exam habit: write the two design equations, the $F_{A0}/(-r_A)$ table, the rectangle-vs-area argument, and the Simpson arithmetic. The deck's own printed answers are exactly 6.4 and 2.165 m³.

$$V_{\text{CSTR}} = 6.4 \text{ m}^3, V_{\text{PFR}} = 2.165 \text{ m}^3 \text{ (the PFR is 4.235 m}^3 \text{ smaller).}$$

QQ4 — Two CSTRs in series. Ch2, slides 31–36

Source: Ch2-krd (1).pdf, slides 35–36.

For two CSTRs in series, 40% conversion is achieved in the first reactor. What is the volume of each reactor necessary to achieve 80% overall conversion of the entering species? (Same feed and rate data as Q3.)

The design equation for the n -th CSTR (slides 32–34). Balance A over reactor 1: In – Out + Generation = 0, i.e. $F_{A0} - F_{A1} + r_{A1}V_1 = 0$; with $F_{A1} = F_{A0} - F_{A0}X_1$ this gives $V_1 = F_{A0} \left(\frac{1}{-r_{A1}} \right) X_1$. Over reactor 2 the balance is $F_{A1} - F_{A2} + r_{A2}V_2 = 0$; substituting $F_{A1} = F_{A0}(1 - X_1)$ and $F_{A2} = F_{A0}(1 - X_2)$ leaves the difference of conversions:

$$V_2 = \frac{F_{A1} - F_{A2}}{-r_{A2}} = \frac{F_{A0}}{-r_{A2}} (X_2 - X_1).$$

Graphically each CSTR is a rectangle whose height is its *own exit* value of $F_{A0}/(-r_A)$.

Step 1 — reactor 1 ($X_1 = 0.4$, slide 36). $F_{A0}/(-r_A)|_{0.4} = 2.05$ m³:

$$V_1 = 2.05 \text{ m}^3 \times 0.4 = 0.82 \text{ m}^3.$$

Step 2 — reactor 2 ($X_1 = 0.4 \rightarrow X_2 = 0.8$). $F_{A0}/(-r_A)|_{0.8} = 8.00$ m³:

$$V_2 = 8.00 \text{ m}^3 \times (0.8 - 0.4) = 3.20 \text{ m}^3.$$

Step 3 — total and comparison. $V = V_1 + V_2 = 0.82 + 3.20 = 4.02$ m³, against 6.4 m³ for a single CSTR to 80%: splitting the conversion across two tanks saves 2.38 m³, because the first tank never has to operate at the slow $X = 0.8$ rate.

Exam habit: sketch the two Levenspiel rectangles (short one on $0 \leq X \leq 0.4$, tall one on $0.4 \leq X \leq 0.8$) and label the heights 2.05 and 8.00 m³ — the picture carries half the marks.

$$V_1 = 0.82 \text{ m}^3, V_2 = 3.20 \text{ m}^3, \text{ total } 4.02 \text{ m}^3 \text{ (vs } 6.4 \text{ m}^3 \text{ for one CSTR).}$$

QQ5 — Two PFRs in series. Ch2, slides 37–39

Source: Ch2-krd (1).pdf, slides 37–39.

Calculate the reactor volumes V_1 and V_2 for the plug-flow sequence shown when the intermediate conversion is 40% and the final conversion is 80% (same feed as Q3).

Why each PFR integrates over its own slice. Reactor 1 raises X from 0 to 0.4 and reactor 2 from 0.4 to 0.8; each integrates $F_{A0}/(-r_A)$ over that interval, so the two areas simply tile the area of one PFR (slide 44 proves that N PFRs in series give the same conversion as a single PFR of the total volume).

Step 1 — reactor 1 (slide 38), three-point Simpson, $h = 0.2$:

$$V_1 = \int_0^{0.4} \frac{F_{A0}}{-r_A} dX = \frac{0.2}{3} [0.89 + 4(1.33) + 2.05] = \frac{0.2}{3} (8.26) = 0.551 \text{ m}^3.$$

Step 2 — reactor 2 (slide 39), points 0.4, 0.6, 0.8:

$$V_2 = \int_{0.4}^{0.8} \frac{F_{A0}}{-r_A} dX = \frac{0.2}{3} [2.05 + 4(3.54) + 8.00] = \frac{0.2}{3} (24.21) = 1.614 \text{ m}^3.$$

Step 3 — total. $V = V_1 + V_2 = 0.551 + 1.614 = 2.165 \text{ m}^3$ — exactly the single-PFR volume of Q3, which is the point the deck is making: a PFR does not care where (or whether) you split it.

$V_1 = 0.551 \text{ m}^3$, $V_2 = 1.614 \text{ m}^3$, total $2.165 \text{ m}^3 =$ one PFR of Q3.

QQ6 – Mixed reactor, unknown stoichiometry. Ch2, slides 41–42

Source: Ch2-krd (1).pdf, slides 42–43 (the worked mixed-reactor problem).

One liter per minute of liquid containing A and B ($C_{A0} = 0.10$, $C_{B0} = 0.01$ mol/liter) flows into a mixed reactor of volume $V = 1$ liter. The materials react in a complex manner for which the stoichiometry is unknown. The outlet stream contains A, B and C with $C_A = 0.02$, $C_B = 0.03$, $C_C = 0.04$ mol/liter. Find the rate of reaction of A, B and C for the conditions within the reactor.

Why one balance per species. A CSTR at steady state balances each species on its own: In – Out + Generation = 0, $v_0 C_{i0} - v_0 C_i + r_i V = 0$, so with the space time $\tau = V/v_0 = 1/1 = 1$ min

$$r_i = \frac{C_i - C_{i0}}{\tau}.$$

Because the stream is liquid (constant volume), every species can be balanced independently — which is exactly why the question can be solved even though the stoichiometry is unknown.

Step 1 — evaluate the three balances.

$$r_A = \frac{0.02 - 0.10}{1} = -0.08, \quad r_B = \frac{0.03 - 0.01}{1} = +0.02, \quad r_C = \frac{0.04 - 0}{1} = +0.04 \text{ mol}/(\text{L} \cdot \text{min}).$$

A is consumed; B and C are produced.

Step 2 — the ratios reveal the stoichiometry. Rates of reaction are proportional to the stoichiometric coefficients (the relative-rate rule of Part C): $-r_A : r_B : r_C = 0.08 : 0.02 : 0.04 = 4 : 1 : 2$, so the equation consistent with the measurements is $4A \rightarrow B + 2C$.

Exam phrasing: “the reactor is a CSTR, hence each species obeys $r_i = (C_i - C_{i0})/\tau$; the signs give consumption/production and the ratio of the rates gives the stoichiometric coefficients.”

$-r_A = 0.08$, $r_B = 0.02$, $r_C = 0.04$ mol/(L · min); the data imply $4A \rightarrow B + 2C$.

QQ7 – PFRs in two parallel branches. Ch2, slides 45–46

Source: Ch2-krd (1).pdf, slides 45–46.

The reactor setup consists of three plug-flow reactors in two parallel branches: branch D has a reactor of volume 50 liters followed by a reactor of volume 30 liters; branch E has a reactor of volume 40 liters. What fraction of the feed should go to branch D?

The rule (slide 45). PFRs in any parallel / parallel-series combination behave as a single PFR of the total volume *only if* the feed is distributed so that streams which meet have the same composition — i.e. the space time V/F must be the same for each parallel line; “any other way of feeding is less efficient”.

Step 1 — branch volumes. Series PFRs add (slide 44): branch D is $50 + 30 = 80$ L; branch E is 40 L.

Step 2 — equal space times. $V_D/F_D = V_E/F_E \Rightarrow F_D/F_E = 80/40 = 2$.

Step 3 — the fraction.

$$\frac{F_D}{F} = \frac{80}{80 + 40} = \frac{2}{3} \approx 0.67.$$

An earlier revision of these notes misread the figure as $20 + 30$ L and printed $5/9$; the slide prints 50 then 30 liters, so two-thirds is the value consistent with the deck.

$2/3 \approx 0.67$ of the feed to branch D (equal space time $80/80 = 40/40 = 1$ L per unit flow).

QQ8 – P2-7: adiabatic $A + B \rightarrow C$, parts (a)–(e). Ch2, slide 59

Source: Ch2-krd (1).pdf, slides 59–60 (“P2-7”).

The exothermic reaction $A \rightarrow B + C$ was carried out adiabatically and the following data were recorded; the entering molar flow rate of A was $F_{A0} = 300$ mol/min:

X	0	0.2	0.4	0.45	0.5	0.6	0.8	0.9
$-r_A$ (mol/dm ³ ·min)	1.0	1.67	5.0	5.0	5.0	5.0	1.25	0.91

(a) PFR and CSTR volumes for 40%; (b) range of X where the two volumes are identical; (c) maximum X in a 105 dm³ CSTR; (d) X for a 72 dm³ PFR followed by a 24 dm³ CSTR; (e) X for the reverse order.

Step 0 — the Levenspiel column and its shape. $g(X) = F_{A0}/(-r_A) = 300, 179.6, 60, 60, 60, 60, 240, 329.7$ dm³ at the tabulated X . Plot it (slide 60): a V-shaped curve — adiabatic heating makes the rate *rise* at first ($1/(-r_A)$ falls to the flat bottom 60), then reactant depletion makes it climb steeply. All five parts are just rectangles (CSTR: $V = X g(X)$) and areas (PFR: $V = \int_0^X g dX$) on this curve.

(a) 40% conversion. CSTR rectangle at the exit height: $V = 0.4 \times 60 = 24$ dm³. PFR area by trapezia over 0, 0.2, 0.4: $V = 0.1(300 + 2 \times 179.6 + 60) = 71.9$ dm³ — here the PFR is *larger*, because the rate is slow at the entrance.

(b) Where CSTR = PFR. Track both with X : at $X = 0.6$, CSTR = $0.6 \times 60 = 36$ vs PFR = $71.9 + 0.2 \times 60 = 83.9$; at $X = 0.8$, CSTR = $0.8 \times 240 = 192$ vs PFR = $83.9 + \frac{0.2}{2}(60 + 240) = 113.9$. The rectangle overtakes the area between 0.6 and 0.8, at $X \approx 0.69$; below it the CSTR is the smaller reactor, above it the PFR is.

(c) 105 dm³ CSTR. Solve $X g(X) = 105$: $X = 0.6 \Rightarrow 36$, $X = 0.8 \Rightarrow 192$, interpolation on the rising limb gives $X \approx 0.70$.

(d) PFR then CSTR. The PFR area reaches 72 dm³ essentially at $X_1 = 0.40$ (part (a): 71.9). The CSTR then needs $(X_2 - X_1) g(X_2) = 24$: on the bottom/rising limb, $X_2 = 0.64$ gives $0.24 \times g(0.64) \approx 0.24 \times 96 \approx 23$ — so $X_2 \approx 0.64$.

(e) CSTR then PFR. $X_1 g(X_1) = 24$ has its useful root at $X_1 = 0.40$ (the height 60 rectangle). The PFR must add 72 dm³ from 0.40: the area $0.4 \rightarrow 0.6$ is 12, $0.6 \rightarrow 0.8$ is 15, $0.8 \rightarrow 0.9$ is 28.5 — 55.5 at the last datum, so the remaining area is eaten just beyond $X = 0.9$ where the data end; quote $X_2 \approx 0.90$.

The V-shaped curve is the exam point: at low X the CSTR wins (it sits on the flat bottom while the PFR drags through the slow entrance); past $X \approx 0.69$ the PFR wins. Order the units accordingly (slide 58: best arrangement rules).

(a) 24 dm³ / 71.9 dm³; (b) about $X = 0.69$; (c) $X \approx 0.70$; (d) $X \approx 0.64$; (e) $X \approx 0.90$.

Part C – Rate laws and stoichiometry (Ch3-krd.pdf)**QQ9 – Relative rates: NO oxidation.** Ch3, slides 5–7

Source: Ch3-krd.pdf, slides 5–7 (“Example”).

$2\text{NO} + \text{O}_2 \rightarrow 2\text{NO}_2$; NO_2 forms at 4 mol/(m³·s).

$\frac{r_{\text{NO}}}{-2} = \frac{r_{\text{O}_2}}{-1} = \frac{r_{\text{NO}_2}}{2}$ gives $r_{\text{NO}} = -4$ and $r_{\text{O}_2} = -2$ mol/(m³·s), exactly as on the deck’s solution slide.

$r_{\text{NO}} = -4$, $r_{\text{O}_2} = -2$ mol/(m³·s).

QQ10 – Relative rates: $2A + 3B \rightarrow 5C$. Ch3, slides 6–7

Source: Ch3-krd.pdf, slides 6–7 (“Exercise”).

$-r_A = 10$ mol/(dm³·s); find the rates of B and C.

$\frac{r_A}{-2} = \frac{r_B}{-3} = \frac{r_C}{5}$: $r_B = \frac{-3}{-2}(-10) = -15$; $r_C = \frac{5}{-2}(-10) = +25$ mol/(dm³·s) (deck values).

$r_B = -15$, $r_C = +25$ mol/(dm³·s).

QQ11 – Saponification stoichiometric table. Ch3, slides 42–44

Source: Ch3-krd.pdf, slides 42–44 (“Example”).

$3\text{NaOH} + (\text{C}_{17}\text{H}_{35}\text{COO})_3\text{C}_3\text{H}_5 \rightarrow 3\text{C}_{17}\text{H}_{35}\text{COONa} + \text{C}_3\text{H}_5(\text{OH})_3$, i.e. $3A + B \rightarrow 3C + D$, liquid phase, X = conversion of NaOH.

$\Theta_B = \frac{1}{3}$, $\Theta_C = \Theta_D = 0$ (as fed): $C_A = C_{A0}(1 - X)$; $C_B = C_{A0}(\Theta_B - \frac{X}{3})$; $C_C = C_{A0}X$; $C_D = C_{A0}\frac{X}{3}$ — the deck’s table (slides 43–44), with $V = V_0$ because the mixture is liquid.

Concentrations as above; e.g. with $\Theta_B = N_{B0}/N_{A0}$ the B entry is $C_{A0}(\Theta_B - X/3)$.

QQ12 – Flow-system table for $A + 2B \rightarrow C + D$. Ch3, slides 48–51

Source: Ch3-krd.pdf, slides 48–51 (“Example”).

Liquid phase, $C_{A0} = 1.8$, $C_{B0} = 6.6 \text{ kmol/m}^3$; A is the basis.

$\Theta_B = 6.6/1.8 = 3.67$. Effluent rates: $F_A = F_{A0}(1 - X)$, $F_B = F_{A0}(3.67 - 2X)$, $F_C = F_{A0}X$, $F_D = F_{A0}X$; concentrations $C_i = F_i/v_0$, e.g. $C_B = C_{A0}(3.67 - 2X)$ (deck slides 49–51).

$C_A = C_{A0}(1 - X)$, $C_B = C_{A0}(3.67 - 2X)$, $C_C = C_D = C_{A0}X$.

QQ13 – Dimerisation $2A \rightarrow R$ in a MFR. Ch3, slide 61

Source: Ch3-krd.pdf, slide 61; repeated in Ch5-krd.pdf, slide 77 (“L 5.2”).

Pure gaseous A ($C_{A0} = 100 \text{ mmol/L}$) dimerises in a MFR ($V = 0.1 \text{ L}$). Runs: $v_0 = 10.0, 3.0, 1.2, 0.5 \text{ L/h}$ with $C_A = 85.7, 66.7, 50, 33.4 \text{ mmol/L}$. Find a rate equation.

Gas phase, $\varepsilon_A = \frac{1-2}{2} = -0.5$: $C_A = C_{A0}(1 - X)/(1 - 0.5X)$ gives $X = 0.250, 0.500, 0.667, 0.800$. MFR design: $-r_A = F_{A0}X/V = v_0C_{A0}X/V$: $-r_A = 2502, 1499, 800, 400 \text{ mmol/(L}\cdot\text{h)}$. Log-log slope of $(-r_A)$ vs C_A is $1.96 \approx 2$; $-r_A/C_A^2 = 0.34, 0.34, 0.32, 0.36$.

$-r_A \approx 0.34 C_A^2$ with C_A in mmol/L and $-r_A$ in mmol/(L·h) (second order in A).

QQ14 – Phosphine cracking PFR. Ch3, slide 63

Source: Ch3-krd.pdf, slide 63 (“Example”).

$4\text{PH}_3 \rightarrow \text{P}_4 + 6\text{H}_2$ at 649 C, first order with $k = 10 \text{ h}^{-1}$. Size a PFR at 649 C and 460 kPa for 80% conversion of 40 mol/h pure PH_3 .

$C_{A0} = P/RT = 460\,000/(8.314 \times 922.15) = 60.0 \text{ mol/m}^3$; $\varepsilon_A = (10 - 4)/4 = 1.5$. $V = \frac{F_{A0}}{kC_{A0}} \left[(1 + \varepsilon) \ln \frac{1}{1-X} - \varepsilon X \right] = \frac{40}{10 \times 60} [2.5 \ln 5 - 1.2] = 0.0667(2.419 + 0.402 - 0.402) = 0.188 \text{ m}^3$.

$V \approx 0.19 \text{ m}^3$ (about 190 L).

Part D – Isothermal design (Ch4_Krd.pdf)**QQ15 – Ethylene-oxide batch data.** Ch4, slides 17–20

Source: Ch4_Krd.pdf, slides 17–20.

500 mL of 2 M EO mixed with 500 mL of water (0.9 wt% H_2SO_4 catalyst) at 55 C ($C_{A0} = 1.0 \text{ kmol/m}^3$ after mixing); glycol concentrations recorded vs time. Determine k ; and the batch time for 90% conversion.

$\ln(C_{A0}/C_A)$ vs t is linear; the deck’s slope is $k = 0.311 \text{ min}^{-1}$ (our least-squares fit of the printed $\ln$ column gives 0.315). Then $t_{0.9} = \ln(10)/k = 7.4 \text{ min}$. The deck’s scaling slide: with $k = 10^{-4} \text{ s}^{-1}$, $t_{0.9} = 2.3/10^{-4} = 23\,000 \text{ s} = 6.4 \text{ h}$.

$k \approx 0.311 \text{ min}^{-1}$; $t_{90} \approx 7.4 \text{ min}$ (and 6.4 h for $k = 10^{-4} \text{ s}^{-1}$).

QQ16 – Ethylene glycol CSTR. Ch4, slides 29–38

Source: Ch4_Krd.pdf, slides 29–38.

Produce 200 million lb/yr of EG at 80% conversion; 1 lbmol/ft³ EO solution mixed with an equal volume of water (0.9 wt% H_2SO_4); $k = 0.311 \text{ min}^{-1}$. (a) CSTR volume; (b) conversion of two 800-gal reactors in parallel; (c) in series.

$F_C = 200 \times 10^6/62 \text{ lbmol/yr} = 3.226 \times 10^6/\text{yr}$; $F_{A0} = F_C/0.8 = 7.67 \text{ lbmol/min}$; $v_{A0} = 7.67 \text{ ft}^3/\text{min}$, $v_0 = 15.34 \text{ ft}^3/\text{min}$, $C_{A0} = 0.5 \text{ lbmol/ft}^3$. (a) $V = \frac{F_{A0}X}{kC_{A0}(1-X)} = \frac{7.67(0.8)}{0.311(0.5)(0.2)} = 197.3 \text{ ft}^3$ (deck value). (b) Each reactor sees v_{A0} : $\tau = \frac{800/7.48}{7.67} = 13.94 \text{ min}$, $Da = k\tau = 4.34$, $X = \frac{Da}{1+Da} = 0.81$. (c) Each sees v_0 : $\tau = 6.97 \text{ min}$, $Da = 2.17$, $X = 1 - \frac{1}{(1+Da)^2} = 0.90$ (deck values).

(a) 197.3 ft³; (b) $X \approx 0.81$; (c) $X \approx 0.90$.

QQ17 – Ethane cracking PFR and tube count. Ch4, slides 44–45

Source: Ch4_Krd.pdf, slides 44–45.

300 million lb/yr ethylene from pure ethane, irreversible elementary rate law, 80% conversion, 1100 K, 6 atm; $k = 0.072 \text{ s}^{-1}$ at 1000 K, $E = 82 \text{ kcal/mol}$. How many 0.0205 ft³, 40-ft PFR tubes in parallel?

$k(1100) = 0.072 \exp\left[\frac{82000}{1.987} \left(\frac{1}{1000} - \frac{1}{1100}\right)\right] = 3.07 \text{ s}^{-1}$; $C_{A0} = 6 \times 101325/(8.314 \times 1100) = 66.5 \text{ mol/m}^3$; $F_{A0} = 300 \times 10^6/28 \text{ lbmol/yr} = 154 \text{ mol/s}$; $v_0 = 2.32 \text{ m}^3/\text{s}$. $\text{C}_2\text{H}_6 \rightarrow \text{C}_2\text{H}_4 + \text{H}_2$; $\varepsilon = 1$; $V = \frac{v_0}{k} \left[2 \ln \frac{1}{1-X} - X\right] = \frac{2.32}{3.07} (2.419) = 1.83 \text{ m}^3 = 64.6 \text{ ft}^3$. Tubes = $64.6/0.0205 \approx 3.2 \times 10^3$.

$V \approx 1.8 \text{ m}^3$ (65 ft³); about 3150 tubes of 0.0205 ft³ in parallel.

The deck solves the same problem numerically in MATLAB (slides 84–95) without printing the final volume; the closed-form first-order gas-phase integral above is the method the deck derives on slides 42–43.

QQ18 – PBR with pressure drop, $A + B \rightarrow 2C$. Ch4, slides 55–58

Source: Ch4_Krd.pdf, slides 55–58.

Equimolar feed, $k = 1.5 \text{ dm}^6/(\text{mol} \cdot \text{kg} \cdot \text{min})$, $\alpha = 0.0099 \text{ kg}^{-1}$, $\varepsilon = 0$. Find X at $W = 100 \text{ kg}$.

$\frac{dX}{dW} = \frac{kC_{A0}^2}{F_{A0}} (1-X)^2(1-\alpha W)$ integrates to $\frac{X}{1-X} = \frac{kC_{A0}^2}{F_{A0}} \left(W - \frac{\alpha W^2}{2}\right)$, $p = (1-\alpha W)^{1/2}$. The deck prints $X = 0.75$ for $\alpha = 0$, which fixes the group $\frac{kC_{A0}^2}{F_{A0}} = \frac{3}{100} = 0.03 \text{ kg}^{-1}$; then $\frac{X}{1-X} = 0.03(100 - 49.5) = 1.515 \Rightarrow X = 0.60$ — the deck's two answers.

$X = 0.60$ with pressure drop, 0.75 without.

The slide does not print C_{A0} and F_{A0} separately; the combined group 0.03 kg^{-1} is implied by its own $\alpha = 0$ answer and is stated here instead of being invented.

QQ19 – PBR + CSTR in series. Ch4, slides 76–77

Source: Ch4_Krd.pdf, slides 76–77.

First order in p_A , $A \rightarrow B$ (gas), PBR with 50 kg catalyst gives 50% conversion at $P_0 = 20 \text{ atm}$; $\alpha = 0.018 \text{ kg}^{-1}$. A CSTR with the same 50 kg is added in series. (a) Which order gives the higher conversion? (b) The exit conversion; (c) pressure at the PBR exit; (d) effect of halving catalyst diameter and enlarging the pipe 50% (turbulent).

Shooting the coupled ODEs $\frac{dX}{dW} = a \frac{1-X}{1+X} p$, $\frac{dp}{dW} = -\frac{\alpha}{2p}$ to match $X(50) = 0.5$ gives $a = kP_0/F_{A0} = 0.0247 \text{ kg}^{-1}$. (c) $p = \sqrt{1-\alpha W} = \sqrt{0.1} = 0.316 \Rightarrow P = 6.3 \text{ atm}$. (b) PBR→CSTR: $X = 0.60$; CSTR→PBR: $X = 0.69$ — putting the CSTR first is better, because the PBR then runs at full pressure. (d) Turbulent $\alpha \propto D_p^{-2} D_{\text{pipe}}^{-4}$; $\alpha' = 0.018 \times 2 \times 1.5^{-4} = 0.0071$, and PBR→CSTR rises to $X \approx 0.73$ (similarly higher in the other order).

(a) CSTR upstream; (b) $X \approx 0.69$ (0.60 the other way); (c) 6.3 atm; (d) $X \approx 0.73$.

QQ20 – Series reactions in a batch reactor. Ch4, slides 78–79

Source: Ch4_Krd.pdf, slides 78–79.

$A \xrightarrow{k_1} B \xrightarrow{k_2} C$, $k_1 = 1 \text{ h}^{-1}$, $k_2 = 2 \text{ h}^{-1}$, $C_A(0) = 5 \text{ mol/m}^3$, $C_B(0) = C_C(0) = 0$.

$C_A = 5e^{-t}$; $C_B = 5 \frac{k_1}{k_2 - k_1} (e^{-k_1 t} - e^{-k_2 t}) = 5(e^{-t} - e^{-2t})$; $C_C = 5 - C_A - C_B$. C_B peaks at $t_{\max} = \frac{\ln(k_2/k_1)}{k_2 - k_1} = 0.693 \text{ h}$ with $C_B^{\max} = 1.25 \text{ mol/m}^3$.

$t_{\max} = \ln 2 \approx 0.69 \text{ h}$, $C_B^{\max} = 1.25 \text{ mol/m}^3$; profiles as above.

QQ21 – PFR pressure drop: pipe length. Ch4, slides 90–98

Source: Ch4_Krd.pdf, slides 90–98.

$A \rightarrow B$, first order, $P_0 = 10 \text{ atm}$, $Q_0 = 0.005 \text{ m}^3/\text{s}$, 300 K, $k = 0.1 \text{ s}^{-1}$, $M = 60 \text{ g/mol}$, $\mu = 10^{-5} \text{ Pa} \cdot \text{s}$. Length for (a) 2 cm pipe, 10%; (b) 1.5 cm pipe, 10%; (c) 1.5 cm pipe, 20%?

$\rho_0 = P_0 M/RT = 24.4 \text{ kg/m}^3$, $v_0 = Q_0/A = 15.9 \text{ m/s}$ (2 cm), $F_{A0} = 2.03 \text{ mol/s}$. Without drop, $L_{10} = \frac{v_0}{k} \ln \frac{1}{0.9} = 16.8 \text{ m}$ and $L_{20} = 35.5 \text{ m}$ (2 cm). The deck integrates the coupled dX/dz , dP/dz ODEs (slides 92–95) and reports (a) $\approx 20 \text{ m}$ for 10% in the 2 cm pipe; (b) 10% still possible in the 1.5 cm pipe ($\approx$ longer, steeper pressure fall); (c) 20% *not* possible — the pressure collapses before 20% is reached (slide 98: “10% conversion is possible, but 20% is not”).

(a) ≈ 20 m; (b) 10% achievable; (c) 20% unattainable in the 1.5 cm pipe.

Part E – Collection and analysis of rate data (Ch5_-krd.pdf)

QQ22 – Trityl + methanol. Ch5, slides 12–17

Source: Ch5_-krd.pdf, slides 12–17 and 42–49.

$(\text{C}_6\text{H}_5)_3\text{CCl} + \text{CH}_3\text{OH} \rightarrow (\text{C}_6\text{H}_5)_3\text{COCH}_3 + \text{HCl}$; $C_{B0} = 0.5$ M; trityl data ($\times 10^3$ mol/dm³): 50, 38, 30.6, 25.6, 22.2, 19.5, 17.4 at $t = 0, 50, \dots, 300$ min. (1) order in trityl; (2) rate constant given first order in methanol.

Finite-difference rates $-dC_A/dt$ (deck slide 17): 2.86, 1.94, 1.24, 0.84, 0.61, 0.48, 0.36×10^{-4} . Log-log slope = $1.99 \approx 2$ (deck) — second order in trityl. Integral check: $1/C_A - 1/C_{A0}$ vs t is linear with slope $k' = 0.124 \approx 0.125$ dm³/(mol·min). With $-r_A = kC_A^2C_B$ and $C_B \approx C_{B0}$: $k = k'/0.5 = 0.25$ dm⁶/(mol²·min). The deck's nonlinear regression returns $\alpha \approx 2$ with the same k' .

Second order in trityl; $k' = 0.125$ dm³/(mol·min), $k = 0.25$ dm⁶/(mol²·min).

QQ23 – Dimethyl ether decomposition bomb. Ch5, slide 75

Source: Ch5_-krd.pdf, slide 75 (“F 5.8”).

$(\text{CH}_3)_2\text{O} \rightarrow \text{CH}_4 + \text{H}_2 + \text{CO}$ at 504 C, constant volume; π (mmHg) = 408, 488, 562, 799 at $t = 390, 777, 1195, 3155$ s and 931 at $t = \infty$. (a) Why is π_0 missing, and estimate it; (b) order and rate constant.

(a) The bomb was already hot and reacting before the first reading; since 1 mol $\rightarrow$ 3 mol, $\pi_0 = \pi_\infty/3 = 310$ mmHg. (b) $p_A = (\pi_\infty - \pi)/2 = 261.5, 221.5, 184.5, 66.0$ mmHg; the four values of $\ln(p_{A0}/p_A)/t$ are 4.4, 4.3, 4.3, 4.9×10^{-4} s⁻¹ — first order.

$\pi_0 \approx 310$ mmHg; first order, $k \approx 4.4 \times 10^{-4}$ s⁻¹.

QQ24 – First-order reversible $A \rightleftharpoons R$. Ch5, slide 72

Source: Ch5_-krd.pdf, slide 72.

$C_{A0} = 0.5$ mol/L, $C_{R0} = 0$; after 8 min $X = 33.3\%$; $X_e = 66.7\%$. Find the rate equation.

$\ln \frac{X_e}{X_e - X} = (k_1 + k_2)t$: $\ln 2 = (k_1 + k_2)(8) \Rightarrow k_1 + k_2 = 0.0866$ min⁻¹; $K_e = X_e/(1 - X_e) = 2 = k_1/k_2$. Hence $k_1 = 0.0578$, $k_2 = 0.0289$ min⁻¹ and $-r_A = 0.0578 C_A - 0.0289 C_R$ (mol/L·min).

$-r_A = 0.058 C_A - 0.029 C_R$ mol/(L·min).

QQ25 – Half-order batch; $2A \rightarrow B$ volume change. Ch5, slide 73

Source: Ch5_-krd.pdf, slide 73.

(i) $A \rightarrow R$, $-r_A = 3C_A^{0.5}$ mol/(L·h), $C_{A0} = 1$ mol/L: conversion after 1 h? (ii) Gas $2A \rightarrow B$, 80% A, constant pressure: volume falls 20% in 3 min; first-order k ?

(i) $2(\sqrt{C_{A0}} - \sqrt{C_A}) = 3t$ gives $\sqrt{C_A} < 0$ for $t > 0.667$ h: the charge is consumed in 40 min, so $X(1 \text{ h}) = 1$. (ii) $V/V_0 = 1 + \varepsilon_A X$ with $\varepsilon_A = 0.8 \times \frac{1-2}{2} = -0.4$; $0.8 = 1 - 0.4X \Rightarrow X = 0.5$; $k = \ln \frac{1}{1-0.5}/3 = 0.231$ min⁻¹.

(i) $X = 1$ (complete at 40 min); (ii) $k = 0.231$ min⁻¹.

QQ26 – $A \rightarrow 3R$ in a MFR (L 5.19). Ch5, slide 74

Source: Ch5_-krd.pdf, slide 74.

Pure A at 3 atm, 30 C ($C_{A0} = 120$ mmol/L) into a 1-L MFR; $v_0 = 0.06, 0.48, 1.5, 8.1$ L/min with $C_A = 30, 60, 80, 105$ mmol/L.

$\varepsilon_A = 2$: $X = \frac{C_{A0} - C_A}{C_{A0} + 2C_A} = 0.50, 0.25, 0.143, 0.045$; $-r_A = v_0 C_{A0} X/V = 3.6, 14.4, 25.7, 44.0$ mmol/(L·min). Log-log slope = 2.00; $-r_A/C_A^2 = 0.0040$ in all four runs.

$-r_A = 0.004 C_A^2$ (C_A in mmol/L, $-r_A$ in mmol/(L·min)), i.e. 4 L/(mol·min) in molar units.

QQ27 – $A \rightarrow 2.5R$ batch, pressure data (L 3.25). Ch5, slide 76

Source: Ch5.-krd.pdf, slide 76.

0 C, constant volume, pure A: p_A (mm) = 760, 600, 475, 380, 320, 275, 240, 215, 150 at $t = 0, 2, \dots, 14, \infty$.

$p_{A,\infty} = 150 \neq 0$: the reaction approaches equilibrium. First order towards it: $\ln \frac{p_{A0} - p_{Ae}}{p_A - p_{Ae}} = kt$ gives $k = 0.152\text{--}0.162$ at every point (fit 0.160 min^{-1}) — a first-order reversible rate equation represents the data.

First-order approach to $p_{Ae} = 150$ mm with $k \approx 0.16 \text{ min}^{-1}$.

QQ28 – Reaction bomb $2A \rightarrow B$ at 100 C. Ch5, slides 78–79

Source: Ch5.-krd.pdf, slides 78–79.

Pure A at 1 atm and 25 C is jumped to 100 C; π falls from 1.14 atm at $t = 1$ min to 0.728 atm at $t = 20$ min (twelve readings, e.g. 1.04, 0.982, 0.940, 0.905, 0.870 at $t = 2..6$); after the weekend no A remains. Rate equation in mol, L, min.

$2A \rightarrow B$ with $\pi_0(100 \text{ C}) = 1$ atm: $p_A = 2\pi - 1$. Least squares: second order ($1/p_A$ linear) fits with $r^2 = 0.964$ vs 0.886 first order, giving $k_p = 0.074 \text{ atm}^{-1}\text{min}^{-1}$; with $p_A = C_A RT$ ($RT = 30.6 \text{ L}\cdot\text{atm/mol}$ at 373 K), $k_C = k_p RT \approx 2.3 \text{ L}/(\text{mol}\cdot\text{min})$.

Second order in A; $k \approx 0.074 \text{ atm}^{-1}\text{min}^{-1} \approx 2.3 \text{ L}/(\text{mol}\cdot\text{min})$ at 100 C.

QQ29 – The gambling tables. Ch5, slide 80

Source: Ch5.-krd.pdf, slide 80.

Losses proportional to money held: table 1 halves it in 4 h, table 2 in 2 h. Starting with Rs. 1000 at both tables at once, when is Rs. 10 left?

First-order decay with $k = \ln 2/4 + \ln 2/2 = 0.520 \text{ h}^{-1}$; $t = \ln(1000/10)/k = 4.605/0.520 = 8.9 \text{ h}$.

About 8.9 hours.

QQ30 – Second order: bigger MFR, equal PFR. Ch5, slide 81

Source: Ch5.-krd.pdf, slide 81.

$-r_A = kC_A^2$ gives 50% in a MFR. (i) conversion in a MFR of 6 times the volume; (ii) in a PFR of the same size.

Original: $kC_{A0}\tau = \frac{X}{(1-X)^2} = 2$. (i) $kC_{A0}\tau' = 12 = \frac{X}{(1-X)^2} \Rightarrow X = 0.75$. (ii) PFR: $kC_{A0}\tau = \frac{X}{1-X} = 2 \Rightarrow X = 2/3$.

(i) $X = 0.75$; (ii) $X = 0.67$.

QQ31 – $A + 3B \rightarrow 6R$ with inerts. Ch5, slide 82

Source: Ch5.-krd.pdf, slide 82.

Feed $C_{A0} = 100$, $C_{B0} = 200$, $C_{I0} = 100$; $C_A = 40$ at the exit. Find X_A , X_B , C_B .

$X_A = 0.6$; $X_B = X_A \frac{3C_{A0}}{C_{B0}} \cdot \frac{1}{3} \times 3 = 0.9$; $C_B = 200 - 3(60) = 20 \text{ mol/L}$.

$X_A = 0.6$, $X_B = 0.9$, $C_B = 20$.

QQ32 – Gaseous feed with T and π change. Ch5, slide 83

Source: Ch5.-krd.pdf, slide 83.

$T_0 = 1000 \text{ K}$, $\pi_0 = 5 \text{ atm}$, $C_{A0} = 100$, $C_{B0} = 200$; $A + B \rightarrow 5R$; at the exit $T = 400 \text{ K}$, $\pi = 4 \text{ atm}$, $C_A = 20$. Find X_A , X_B , C_B .

$y_{A0} = 1/3$, $\delta = +3$, $\varepsilon_A = 1$: $C_A = C_{A0} \frac{1-X}{1+X} \frac{T_0}{T} \frac{\pi}{\pi_0} = 32 \frac{1-X}{1+X} = 20 \Rightarrow X_A = 0.231$; $X_B = X_A C_{A0}/C_{B0} = 0.115$; $C_B = 200 \frac{1-X_B}{1+X_A} \times 0.32 = 46.0$.

$X_A = 0.23$, $X_B = 0.12$, $C_B \approx 46$.

QQ33 – Sucrase kinetics, Michaelis–Menten. Ch5, slide 84

Source: Ch5.-krd.pdf, slide 84.

$C_{A0} = 1.0$, $C_{S0} = 0.01$ mmol/L; C_A (mmol/L) = 0.68, 0.60, 0.55, 0.38, 0.27, 0.16, 0.09, 0.04, 0.018, 0.006, 0.0025 at $t = 1..11$ h. Does Michaelis–Menten fit? Evaluate k_3 , C_M .

Integrated MM: $k_3 C_{S0} t = (C_{A0} - C_A) + C_M \ln \frac{C_{A0}}{C_A}$, linearised as $\frac{t}{\ln(C_{A0}/C_A)} = \frac{1}{k_3 C_{S0}} \frac{C_{A0} - C_A}{\ln(C_{A0}/C_A)} + \frac{C_M}{k_3 C_{S0}}$. Least squares over the eleven points: slope = 3.12 h, intercept = 1.61 h, giving $k_3 C_{S0} = 0.321 \text{ h}^{-1}$ ($k_3 \approx 32 \text{ h}^{-1}$) and $C_M \approx 0.52$ mmol/L; the points lie on the line ($r^2 > 0.99$), so the fit is reasonable.

Yes: $k_3 \approx 32 \text{ h}^{-1}$ (at $C_{S0} = 0.01$), $C_M \approx 0.5$ mmol/L.

QQ34 – Reversible PFR: equilibrium then actual conversion. Ch5, slide 61

Source: Ch5.-krd.pdf, slide 61.

A PFR (2 m^3) processes aqueous feed (100 L/min) with $C_{A0} = 100$ mmol/L; $A \rightleftharpoons B$. Find the equilibrium conversion, then the actual conversion (and repeat for a CSTR).

The slide prints no rate constants; the textbook problem it reproduces (Levenspiel, the companion MFR version of which is printed with $-r_A = 0.04C_A - 0.01C_R$ mol/(L·min)) supplies them: $K_e = k_1/k_2 = 4 \Rightarrow X_e = \frac{4}{5} = 0.80$. $-r_A = 0.05(C_A - 0.02)$; $\tau = V/v_0 = 20$ min; $\ln \frac{C_{Af} - 0.02}{0.08} = -(k_1 + k_2)\tau = -1 \Rightarrow C_{Af} = 0.0494$, $X = 0.51$. (CSTR: $0.05(C_A - 0.02) \times 20 = C_{A0} - C_A \Rightarrow C_A = 0.06$, $X = 0.40$.)

$X_e = 0.80$; PFR $X \approx 0.51$; CSTR $X \approx 0.40$ (rate law quoted from the textbook original, as the slide omits it).

QQ35 – $A \rightarrow R + 2S$ from total-pressure data. Ch5, slide 62

Source: Ch5.-krd.pdf, slide 62.

Pure A, constant volume; π (mmHg) = 7.5, 10.5, 12.5, 15.8, 17.9, 19.4 at $t = 0, 2.5, 5, 10, 15, 20$ min.

$\Delta n = +2$: $p_A = \pi_0 - \frac{\pi - \pi_0}{2} = \frac{3\pi_0 - \pi}{2} = 7.5, 6.0, 5.0, 3.35, 2.3, 1.55$. $\ln(p_{A0}/p_A)/t = 0.089, 0.081, 0.081, 0.079, 0.079$ — first order, $k \approx 0.08 \text{ min}^{-1}$.

First order with $k \approx 0.08 \text{ min}^{-1}$.

Part F – Multiple reactions (Ch6_krd.pdf)

QQ36 – Product distribution, three contacting patterns. Ch6, slide 32

Source: Ch6_krd.pdf, slide 32.

Aqueous $A + B \xrightarrow{k_1} R$ (desired), $A + B \xrightarrow{k_2} S$, with $r_R = 1.0 C_A^{1.5} C_B^{0.3}$, $r_S = 1.0 C_A^{0.5} C_B^{1.8}$ mol/(L·min). Equal streams of A and B at 20 mol/L each are fed ($C_{A0} = C_{B0} = 10$ after mixing); 90% conversion of A. Find C_R for (a) plug flow, (b) mixed flow, (c) plug flow with side entrances of B.

With $C_A = C_B = c$: $\varphi(R/A) = \frac{r_R}{r_R + r_S} = \frac{1}{1 + c^{0.5}}$. (a) $C_R = \int_1^{10} \frac{dc}{1 + \sqrt{c}} = 2[u - \ln(1 + u)]_1^{\sqrt{10}} = 2.86 \text{ mol/L}$. (b) $C_R = \varphi|_{c=1} (10 - 1) = 0.5 \times 9 = 4.5 \text{ mol/L}$. (c) Side entry of B holds C_B at the low exit level (= 1): $\varphi = \frac{c}{c+1}$, $C_R = \int_1^{10} \frac{cdc}{c+1} = 9 - \ln \frac{11}{2} = 7.30 \text{ mol/L}$.

(a) 2.86; (b) 4.50; (c) 7.30 mol/L — mixed flow beats plug flow here, and keeping B low is best.

QQ37 – Two MFRs in series. Ch6, slide 33

Source: Ch6_krd.pdf, slide 33.

$r_R = k_1 C_A^2$, $r_S = k_2 C_A$; feed $C_{A0} = 1.0$, $C_{R0} = 0$, $C_{S0} = 0.3$; $\tau_1 = 2.5$, $\tau_2 = 10$ min; first-reactor exit $C_A = 0.4$, $C_R = 0.2$, $C_S = 0.7$. Find the second-reactor exit.

From reactor 1: $k_1 = \frac{0.2/2.5}{0.4^2} = 0.5$; $k_2 = \frac{(0.7-0.3)/2.5}{0.4} = 0.4$ (the S balance checks: $0.4 \times 0.4 = 0.16 = (0.7 - 0.3)/2.5$). Reactor 2: $0.4 - C_A = (0.5 C_A^2 + 0.4 C_A) 10 \Rightarrow C_A = 0.0745$; $C_R = 0.2 + 0.5 C_A^2 (10) = 0.228$; $C_S = 0.7 + 0.4 C_A (10) = 0.998$.

$C_A \approx 0.075$, $C_R \approx 0.23$, $C_S \approx 1.00$ mol/L.

Part G – Non-isothermal design (Ch7_krd.pdf)

QQ38 – Adiabatic PFR: n-butane isomerisation. Ch7, slides 14–17

Source: Ch7-_krd.pdf, slides 14–17.

n -butane $\rightleftharpoons$ isobutane, adiabatic liquid-phase PFR; $k = 31.1 \text{ h}^{-1}$ at 360 K, $E = 65.7 \text{ kJ/mol}$, $\Delta H_{Rx} = -6900 \text{ J/mol}$, $K_C = 3.03$ at 60 C, $C_{A0} = 9.3 \text{ kmol/m}^3$, $C_p = 141/141/161 \text{ J/(mol}\cdot\text{K)}$ (butane/isobutane/pentane). Process 163 kmol/h (90 mol% butane) to $X = 0.70$, feed at 330 K.

Adiabatic line $T = 330 + 43.3X$ (deck); $k(T) = 31.1e^{7902(1/360-1/T)}$; $K_C(T) = 3.03e^{830.3(1/333-1/T)}$; $-r_A = kC_{A0} \left[1 - \left(1 + \frac{1}{K_C}\right)X\right]$; $F_{A0} = 0.9 \times 163 = 146.7 \text{ kmol/h}$. The deck's table (e.g. $-r_A = 39.2$ at $X = 0$, $F_{A0}/-r_A = 23.29 \text{ m}^3$ at 0.7) is reproduced by these expressions; integrating $F_{A0}/(-r_A)$ to $X = 0.7$ gives $V \approx 2.0 \text{ m}^3$ (the deck's coarse Simpson table reads $\approx 2.4 \text{ m}^3$; the value is sensitive near $X_e \approx 0.72$).

$V \approx 2\text{--}2.4 \text{ m}^3$ for 70% (deck table: $F_{A0}/(-r_A) = 23.3 \text{ m}^3$ at $X = 0.7$).

QQ39 – Adiabatic equilibrium with two interstage coolers. Ch7, slides 48–51

Source: Ch7-_krd.pdf, slides 48–51.

$A \rightleftharpoons B$, pure A at 300 K; $K_C(298) = 10^5$, $\Delta H_{Rx} = -20\,000 \text{ cal/mol}$ (as implied by the deck's $K_C(T)$ and adiabatic line), $C_{pA} = C_{pB} = 50 \text{ cal/(mol}\cdot\text{K)}$, so $X_{ad} = 2.5 \times 10^{-3}(T - 300)$. Two coolers return the stream to 350 K; 95% of the equilibrium conversion in each reactor; $F_{A0} = 40 \text{ mol/s}$; cooling water at 270 K, $c_{pC} = 18 \text{ cal/(mol}\cdot\text{K)}$, $U = 100 \text{ cal/(s}\cdot\text{m}^2\cdot\text{K)}$.

$K_C(T) = 10^5 \exp[-33.78(T - 298)/T]$, $X_e = K_C/(1 + K_C)$. Reactor 1: $2.5 \times 10^{-3}(T - 300) = 0.95X_e(T) \Rightarrow T_1 = 459 \text{ K}$, $X_1 = 0.40$. Cool to 350 K. Reactor 2: $X_1 + 2.5 \times 10^{-3}(T - 350) = 0.95X_e(T) \Rightarrow T_2 = 440 \text{ K}$, $X_2 = 0.62$. Duties: $\dot{Q} = \dot{n}C_p(T_{out} - 350)$: $\dot{Q}_1 = 40 \times 50 \times 109 = 218 \text{ kcal/s}$; $\dot{Q}_2 = 40 \times 50 \times 90 = 179 \text{ kcal/s}$ (water flow = $\dot{Q}/(18\Delta T)$ as sized by the deck).

$X \approx 0.40$ after reactor 1 and 0.62 after reactor 2; $\dot{Q}_1 \approx 218$, $\dot{Q}_2 \approx 179 \text{ kcal/s}$.

QQ40 – Propylene-oxide CSTR, adiabatic. Ch7, slides 57–59

Source: Ch7-_krd.pdf, slides 57–59 (and 69–72).

300-gal glass-lined CSTR; $F_{A0} = 43.04 \text{ lbmol/h}$ PO; streams: $46.62 \text{ ft}^3/\text{h}$ PO, $46.62 \text{ ft}^3/\text{h}$ methanol, $233.1 \text{ ft}^3/\text{h}$ water (0.1 wt% H_2SO_4); methanol 71.87 and water 802.8 lbmol/h; $T_0 = 75 \text{ F} = 535 \text{ R}$ after the 17 F heat-of-mixing rise; first order in PO with $k = 16.96 \times 10^{12} e^{-32\,400/RT} \text{ h}^{-1}$; limit 125 F.

$v_0 = 326.3 \text{ ft}^3/\text{h}$, $C_{A0} = 0.132 \text{ lbmol/ft}^3$, $\tau = (300/7.48)/326.3 = 0.123 \text{ h}$. Energy balance: $\sum \Theta_i C_{p,i} \approx 404 \text{ Btu/(lbmol A}\cdot\text{R)}$, $-\Delta H_{Rx} = 36\,400 \text{ Btu/lbmol} \Rightarrow T = 535 + 90.1X$. Solving with $X = \frac{k\tau}{1+k\tau}$: $X = 0.84$, $T = 611 \text{ R} = 151 \text{ F}$ (deck table intersection $\approx 613 \text{ R}$, $X \approx 0.85$). $151 > 125 \text{ F}$: the reactor *cannot* be used adiabatically.

$X \approx 0.84\text{--}0.85$ at $T \approx 611\text{--}613 \text{ R}$ ($\approx 151 \text{ F}$) $> 125 \text{ F}$ — not acceptable.

QQ41 – The same CSTR with a cooling coil. Ch7, slides 60–74

Source: Ch7-_krd.pdf, slides 60–74.

A 40 ft^2 coil, $U = 100 \text{ Btu/(h}\cdot\text{ft}^2\cdot\text{F)}$, coolant at 85 F (545 R). Does the 125 F constraint now hold?

Energy balance adds $UA(T - T_a) = 4000(T - 545) \text{ Btu/h}$: $X = \frac{17\,379(T-535)+4000(T-545)}{43.04 \times 36\,400}$ together with the mole balance. Intersection: $X \approx 0.35\text{--}0.36$, $T \approx 562\text{--}564 \text{ R} = 102\text{--}104 \text{ F}$ — the deck's fsolve gives $X = 0.364$, $T = 563.7 \text{ R}$. The constraint is satisfied with the coil.

Yes: $X \approx 0.36$ at $T \approx 564 \text{ R}$ ($\approx 104 \text{ F}$) $< 125 \text{ F}$.

QQ42 – Parallel reactions in a PFR with heat effects. Ch7, slide 95

Source: Ch7-_krd.pdf, slide 95.

$A \rightarrow B$ ($r_B = k_1 C_A$), $2A \rightarrow C$ ($r_C = k_2 C_A^2$); pure A at 100 mol/s, 150 C, $C_{A0} = 0.1 \text{ mol/dm}^3$; $\Delta H_1 = -20\,000$, $\Delta H_2 = -60\,000 \text{ J/mol A}$; $C_{pA} = C_{pB} = 90$, $C_{pC} = 180 \text{ J/(mol}\cdot\text{C)}$; $E_1/R = 4000 \text{ K}$, $E_2/R = 9000 \text{ K}$; $k_1 = 10$, $k_2 = 0.09$ (as printed, per min). Determine the temperature and flow-rate profiles down the reactor.

Coupled ODEs (per the deck's PFR heat-effects balances): $\frac{dF_A}{dV} = -(r_1 + 2r_2)$, $\frac{dF_C}{dV} = r_2$, $\frac{dT}{dV} = \frac{r_1(-\Delta H_1) + r_2(-\Delta H_2)}{F_A C_{pA} + F_B C_{pB} + F_C C_{pC}}$ with $C_A = F_A/\dot{V}_0$, $\dot{V}_0 = 60\,000 \text{ dm}^3/\text{min}$ (taking the printed rate constants as per minute). Integrating to $V = 10 \text{ m}^3$: $F_A \rightarrow 0$, $F_C \rightarrow 0$ (B dominates), and T rises to $\approx 645 \text{ K}$ — the adiabatic limit for complete conversion to B ($\Delta T = 100 \times 20\,000/(100 \times 90) \approx 222 \text{ K}$ above 423 K).

Both reactions complete within the first m^3 ; $T \rightarrow \approx 645 \text{ K}$, product essentially B.

The slide states the problem and the data but no units for k_1, k_2 beyond “min”; the profile above takes them as printed (per minute). The qualitative result (fast exothermic run to the adiabatic temperature rise) is unaffected.

Part H – Catalysis (Catalysis_Krd.pdf)

QQ43 – Pellet concentration profile. Catalysis, slide 48

Source: Catalysis_Krd.pdf, slide 48 (“Problem”).

$A \rightarrow B$, first order, spherical particle $d_p = 2 \times 10^{-3} \text{ cm}$; $C_{As} = 0.001 \text{ gmol/m}^3$; $D_e = 0.1 \text{ cm}^2/\text{s}$. The concentration halfway to the centre is $1/10$ of C_{As} . (a) C_A at $3 \times 10^{-4} \text{ cm}$ from the external surface; (b) diameter for $\eta = 0.8$.

Sphere, first order: $\frac{C_A}{C_{As}} = \frac{R}{r} \frac{\sinh(\phi r/R)}{\sinh \phi}$. At $r = R/2$: $0.1 = \frac{2 \sinh(\phi/2)}{\sinh \phi} = \frac{1}{\cosh(\phi/2)} \Rightarrow \phi = 2 \operatorname{arcosh}(10) = 5.99$. (a) $r = R - 3 \times 10^{-4} = 7 \times 10^{-4} \text{ cm}$: ratio $= \frac{1}{0.7} \frac{\sinh(0.7 \times 5.99)}{\sinh 5.99} = 0.237$, $C_A = 2.4 \times 10^{-4} \text{ gmol/m}^3$. (b) $\eta = \frac{3}{\phi^2} (\phi \coth \phi - 1) = 0.8 \Rightarrow \phi_{0.8} = 2.04$; $\phi \propto d_p$: $d_{new} = 2 \times 10^{-3} \times \frac{2.04}{5.99} = 6.8 \times 10^{-4} \text{ cm}$.

(a) $C_A \approx 2.4 \times 10^{-4} \text{ gmol/m}^3$; (b) $d_p \approx 6.8 \times 10^{-4} \text{ cm}$.

QQ44 – Internal diffusion: 100 vs 150 C. Catalysis, slide 57

Source: Catalysis_Krd.pdf, slide 57.

First order, $R = 0.2 \text{ cm}$, $D_e = 0.015 \text{ cm}^2/\text{s}$; $k(100 \text{ C}) = 0.93 \text{ s}^{-1}$, $E = 20 \text{ kcal/mol}$; $C_{Ab} = 3.25 \times 10^{-5} \text{ mol/L}$. (a) rate at 100 C ; (b) at 150 C ; (c) apparent activation energy.

(a) $\phi_1 = R\sqrt{k_1/D_e} = 1.575$; $\eta_1 = 0.866$; $-r_A = \eta_1 k_1 C_{Ab} = 2.6 \times 10^{-5} \text{ mol/(L}\cdot\text{s)}$. (b) $k_2 = 0.93 e^{\frac{20,000}{1.987} (1/373 - 1/423)} = 22.5 \text{ s}^{-1}$; $\phi_2 = 7.75$; $\eta_2 = 0.337$; $-r_A = 2.5 \times 10^{-4} \text{ mol/(L}\cdot\text{s)}$. (c) From the two rates, $E_{app} = R \ln(r_2/r_1) / (1/T_1 - 1/T_2) \approx 14.1 \text{ kcal/mol}$ — between E and $E/2$, moving toward the diffusion-limited $E/2 = 10 \text{ kcal/mol}$ as ϕ grows.

(a) 2.6×10^{-5} ; (b) $2.5 \times 10^{-4} \text{ mol/(L}\cdot\text{s)}$; (c) $E_{app} \approx 14 \text{ kcal/mol}$.

Part I – Residence time distributions (Non_ideal_KRD.pdf)

QQ45 – Tracer pulse: $E(t)$ and the 3–6 min fraction. Non-ideal, slides 17–19

Source: Non_ideal_KRD.pdf, slides 17–19 (“Problem”).

$C \text{ (g/m}^3\text{)} = 0, 1, 5, 8, 10, 8, 6, 4, 3, 2.2, 1.5, 0.6, 0$ at $t = 0, 1, \dots, 10, 12, 14 \text{ min}$. Construct $E(t)$ and find the fraction that spent 3–6 min in the reactor.

$\int_0^\infty C dt = 47.4 + 2.6 = 50.0 \text{ g}\cdot\text{min/m}^3$ (Simpson on 0–10 plus 10–14, as on the deck). $E(t) = C(t)/50$: peak $E(4) = 0.20$. Fraction between 3 and 6 min: $\int_3^6 E dt \approx \frac{1}{50} \times \frac{3}{8} (8 + 3 \cdot 10 + 3 \cdot 8 + 6) = 0.465 \approx 0.5$. Mean residence time $t_m = \int t E dt \approx 5.2 \text{ min}$.

$\int C dt = 50 \text{ g}\cdot\text{min/m}^3$; about half the material (0.47–0.51) spent 3–6 min inside; $t_m \approx 5.2 \text{ min}$.

QQ46 – Dispersion model: conversion four ways. Non-ideal, slides 52–59

Source: Non_ideal_KRD.pdf, slides 52–59 (“Problem”).

First order $A \rightarrow B$ in a 10 cm , 6.36 m tubular reactor, $k = 0.25 \text{ min}^{-1}$; tracer test of Q45. Conversion by (i) closed-vessel dispersion model, (ii) PFR, (iii) single CSTR, (iv) tanks in series.

$t_m = 5.13 \text{ min}$, $\sigma^2 = 5.92 \text{ min}^2$ (deck values), $\sigma_\theta^2 = 0.225$; $\sigma_\theta^2 = \frac{2}{Pe} - \frac{2}{Pe^2} (1 - e^{-Pe}) \Rightarrow Pe = 7.7$. $Da = kt_m = 1.28$, $q = \sqrt{1 + 4Da/Pe} = 1.29$: $X = 1 - \frac{4q e^{Pe/2}}{(1+q)^2 e^{Pe q/2} - (1-q)^2 e^{-Pe q/2}} = 0.68$ (deck value). (ii) $X = 1 - e^{-Da} = 0.72$; (iii) $X = \frac{Da}{1+Da} = 0.56$; (iv) $n = t_m^2/\sigma^2 = 4.45$: $X = 1 - (1 + Da/n)^{-n} = 0.68$.

(i) 0.68; (ii) 0.72; (iii) 0.56; (iv) 0.68.

Every number above is recomputed by `krd-answers-check.py` (kept in this folder) from the data quoted here; discrepancies with a deck’s own printed answer, where any, are discussed in that question’s Notes/Warning box.